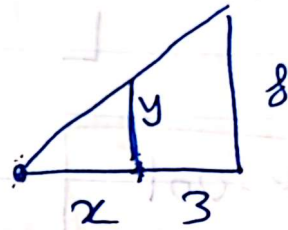
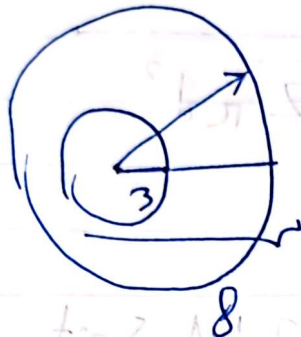


Q1- Hint



$$\frac{x}{3} = \frac{x+3}{8}$$

x = particle displacement

$$V = \frac{\Delta x}{\Delta t} = \frac{x}{\frac{3}{a}} \rightarrow$$

$$a \Delta t = 3$$

$$\Delta t = \frac{3}{a}$$

$$V = \frac{x}{3} a$$

$$M = \frac{x}{3} a$$

Q3 Hint

a)

$$a = \sqrt{\frac{\partial p}{\partial \rho}}$$

$$\rho = \frac{1}{g}$$

E.O.S

write Eq of state in terms of g , then compute $\frac{\partial p}{\partial g}$

b)

$$de = c_v dT \quad \& \quad \boxed{e = c_p T - \frac{a}{v}}$$

$$c_v dT = de = c_p dT + \frac{a}{v^2} dv$$

$$c_v - c_p = \frac{a}{v^2} ?$$

$$\Rightarrow \begin{cases} c_v dT - c_p dT = \frac{a}{v^2} dv \\ c_v - c_p = -\frac{a}{v^2} \frac{dv}{dT} \end{cases}$$

Take derivative from E.O.S